Practice Set 7 Paper 1 – Basic Sciences in Urology

1. Q1. Prune Belly Syndrome. Antenatal diagnosis, pathophysiology and postnatal management.
2. Q2. Lymphatic drainage of testis and principles for designing templates for RPLND.
3. Q3a. Papillary necrosis.
4. Q3b. MAG 3
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Practice paper compiled from the question bank.

Q1. Prune Belly Syndrome. Antenatal diagnosis, pathophysiology and postnatal management.

Type: LAQ (25 marks) (also answered as an SAQ in 2015-Summer Paper 3, Q3f)

Definition

- **PBS (prune belly syndrome)**, also called Eagle–Barrett or triad syndrome: **deficient abdominal wall muscles**, **bilateral intra-abdominal testes** and a **dilated, poorly contracting urinary tract**.
- 1 in 29,000–44,000 live births; **95% male**.
- **Survival depends on renal dysplasia and lung hypoplasia, not on the abdominal wall.**

Antenatal diagnosis

- **Ultrasound** may suggest PBS from 11–14 weeks, but the classic signs are not consistent even at 30 weeks.
- **Findings:** huge bladder filling the abdomen; bilateral hydroureteronephrosis; urachal diverticulum; dilated posterior urethra; echogenic kidneys with cortical cysts (dysplasia); irregular abdominal circumference; fetal ascites; oligohydramnios when severe.
- **Differential diagnosis:** PUV (posterior urethral valves), urethral atresia, megacystis–megaureter.
- Fetal dilatation does **not** predict postnatal renal function.
- **Prenatal management:** intervention has no proven renal benefit. Vesicoamniotic shunting is justified only for **urethral atresia with progressive oligohydramnios**, or to decompress the abdomen and avoid dystocia. Termination is hard to justify except with early, severe oligohydramnios. Plan delivery in a centre with neonatal intensive care.

Pathophysiology

Theory (none proven)	Mechanism
Early urethral obstruction	Back-pressure dilates the tract and stretches the wall; but most have no obstruction after birth
Lateral plate mesoderm defect	This mesoderm forms ureters, bladder, prostate, urethra and gubernaculum, so one defect explains the triad
Intrinsic urinary tract defect	Primary ureteric dilatation and fetal ascites
Yolk sac defect	Primary defect of the yolk sac

- **Genetics:** mostly sporadic, normal karyotype; smooth-muscle genes **MYOCD (myocardin)** and **FLNA (filamin A)** in multiplex families.
- **Kidneys:** dysplasia in about 50% (Potter type II from mesenchymal defect; type IV cystic from obstruction).
- **Ureters:** long and tortuous, worse distally. Collagen replaces smooth muscle, so walls do not coapt and peristalsis fails: **a failed pump, not a blocked pipe**. The result is stasis and infection. **VUR (vesicoureteric reflux) in about 75%.** The proximal ureter has better muscle.
- **Bladder:** large, smooth, collagen-rich; urachal pseudodiverticulum; splayed trigone. Normal compliance but poor contractility, so voiding is "unbalanced", with residual urine.
- **Prostate and urethra:** prostatic hypoplasia dilates the posterior urethra; distal obstruction in about 20%; **megalourethra** (scaphoid: spongiosum deficient; fusiform: all erectile bodies, more severe).

- **Testes:** intra-abdominal; azoospermia; paternity only by sperm retrieval and ICSI (intracytoplasmic sperm injection).
- **Extra-genitourinary (about 75%):** pulmonary hypoplasia; cardiac about 10%; gastrointestinal at least 30% (malrotation); orthopaedic up to 65% (talipes, hip dysplasia).

Classification (Woodard)

Category	Features	Management
I	Oligohydramnios, pulmonary hypoplasia, dysplasia, urethral atresia	Supportive; bladder drainage only
II	Full triad, moderate or unilateral dysplasia	Individualised (controversial group)
III	Mild or incomplete; stable function (majority)	Surveillance

- **Also know:** RUBACE score (Renal, Ureter, Bladder, Abdominal wall, Cryptorchidism, Extra-genitourinary) grades morbidity in survivors; pseudo-prune presents late with renal failure.

Postnatal management

Neonatal (neonatology, nephrology, urology, cardiology):

1. Heart and lungs first: **chest radiograph** for pneumothorax and hypoplasia.
2. Creatinine, BUN (blood urea nitrogen), electrolytes; follow the **trend**, because the first value reflects the mother.
3. Renal and bladder ultrasound once stable.
4. Antibiotic prophylaxis; **circumcision**; strict asepsis.
5. **VCUG (voiding cystourethrogram)** only with renal insufficiency or suspected outlet obstruction.
6. **DMSA (dimercaptosuccinic acid) at 4–6 weeks** for split function; MAG3 (mercaptoacetyltriglycine) for drainage.
7. Outlet obstruction: catheter or suprapubic tube.

Urinary tract:

- **Conservative** for category III: surveillance, treat infection.
- **Surgery for:** recurrent upper tract infection, progressive dilatation or renal decline, high-grade VUR, true obstruction.
- **Temporary diversion:** wide **cutaneous vesicostomy (Blocksom)**. Avoid high ureterostomy, which wastes the good proximal ureter.
- **Reconstruction after 3–6 months** (lung maturity): excise distal ureter, taper and reimplant the healthier proximal ureter; reduction cystoplasty (capacity recurs). **CIC (clean intermittent catheterisation)** or Mitrofanoff for poor emptying.

Orchidopexy: bilateral transabdominal at about 6 months, often with other surgery; later, **staged Fowler–Stephens** (better than single-stage) or laparoscopic.

Abdominoplasty (cosmesis; better cough, voiding, defecation):

Technique	Key feature
Randolph	Transverse; excises worst lower wall
Ehrlich	Vertical; umbilicus kept; vest-over-pants overlap
Monfort	Vertical; umbilicus kept; central fascial plate with lateral flaps overlapped; transperitoneal access for combined surgery

- **Also know:** Furness (extraperitoneal plication); Dénes; corset.

Prognosis and follow-up

- About 20% die perinatally; **up to 30% develop chronic renal failure**. **Nadir creatinine <0.7 mg/dL at 1 year** predicts stable function. Lifelong timed voiding, cultures and imaging; transplant when needed.

Diagram to draw: fetal ultrasound sketch: huge bladder, dilated posterior urethra, bilateral hydroureteronephrosis.

> **Key points to remember:**
> - Triad; 95% male; dysplasia and lungs decide survival.
> - Ureter is a failed pump (collagen); VUR about 75%; keep the proximal ureter.
> - Woodard I–III; vesicostomy for drainage; orchidopexy at about 6 months.
> - Nadir creatinine <0.7 mg/dL is favourable; up to 30% reach renal failure.

Q2. Lymphatic drainage of testis and principles for designing templates for RPLND.

Type: LAQ (25 marks) (lymphatic drainage also answered in 2018-Summer Paper 2, Q2; minimally invasive RPLND in 2025-Winter Paper 3, Q3f)

Embryological basis

- The testis develops at L1–L2 beside the mesonephros and descends, carrying its vessels and lymphatics. So its primary drainage is to the **retroperitoneal (lumbar) nodes near the renal hilum**, not the groin.

Lymphatic pathway

- Lymphatics of the testis and the caput and corpus of the epididymis run in the spermatic cord with the testicular vessels, through the internal ring and along the gonadal vessels to the lumbar nodes.
- Lumbar trunks join at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta), then the **thoracic duct** to the left innominate vein. This explains **left supraclavicular** metastases.
- Lymph flows **cephalad and right to left**.
- **Cauda epididymis** drains with the vas to the **external iliac** nodes.

Nodal region	Extent
Left para-aortic (with preaortic, retroaortic)	Aortic midline to left ureter
Interaortocaval	IVC (inferior vena cava) midline to aortic midline
Right paracaval (with precaval, retrocaval)	IVC midline to right ureter

- **Also know:** ten mapped regions include right and left common iliac, interiliac and right and left suprahilar.

Primary landing zones

	Right testis	Left testis
Primary	**Interaortocaval**	**Left para-aortic** (left hilar)
Secondary	Right paracaval, precaval	Interaortocaval
Crossover	Small but real to left para-aortic	None significant

- **Crossover is only right to left**, matching the overall lymph flow.
- **Donohue mapping (1982):** contralateral and suprahilar spread is rare but rises with bulk; interiliac spread is very rare.
- **Pelvic and inguinal nodes** are involved only when drainage is altered: previous scrotal or inguinal surgery, scrotal violation, or tumour invading the tunica vaginalis. Scrotal skin drains to ipsilateral superficial inguinal nodes. **Hence trans-scrotal orchidectomy is contraindicated.**

Principles for designing RPLND templates

1. **Follow the side-specific landing zones** to remove likely disease and spare the rest.
2. **Standard limits:** renal vessels above, ureters laterally, ureter crossing the common iliac artery below; always remove the gonadal vein and cord stump to the internal ring.
3. **Omit suprahilar and interiliac regions** without bulky disease. This reduces chylous ascites, renal and pancreatic injury, and protects the hypogastric plexus.

4. **Unilateral modified templates** leave contralateral sympathetic fibres untouched. Because of right-to-left crossover, both templates take the para-aortic nodes **above the IMA (inferior mesenteric artery)**.
5. **Nerve sparing:** antegrade ejaculation (emission and bladder-neck closure) depends on **L1–L4 postganglionic sympathetic fibres** (about three each side) joining the superior hypogastric plexus below the IMA. Identify them prospectively before dividing tissue.
6. **Completeness:** split-and-roll, division of lumbar vessels to clear retrocaval and retroaortic tissue, clip lymphatics at the cephalad limit (left renal vein) to prevent chylous ascites.
7. **Extend if needed:** enlarged nodes at operation → full bilateral template (frozen section, Weissbach 1987).

Template	Includes
Right modified	Paracaval, precaval, retrocaval, interaortocaval, right common iliac; para-aortic above IMA
Left modified	Left para-aortic, preaortic, retroaortic, left common iliac; interaortocaval above IMA
Bilateral infrahilar	All three regions both sides, between renal vessels, ureters and ureteric crossing of iliacs
Historical extended	Plus suprahilar and interiliac (thoracoabdominal); most anejaculatory

Template by clinical setting

Setting	Template
Primary RPLND (high-risk stage I or low-volume stage II NSGCT [non-seminomatous germ cell tumour], normal markers)	Modified unilateral nerve-sparing **or** full bilateral infrahilar nerve-sparing; no randomised trial
PC-RPLND (post-chemotherapy), residual mass >1 cm, normal markers	**Full bilateral infrahilar + mass** (standard)
Modified PC template allowed only if	Mass ≤5 cm in landing zone, normal markers, IGCCCG (International Germ Cell Cancer Collaborative Group) good or intermediate risk

- Against unilateral templates: 3–23% of node-positive men have nodes outside the modified template, more with right tumours.

Outcomes and complications

- Primary RPLND: cancer-specific survival about 98.8–100%.
- **Antegrade ejaculation >90%** with templates plus nerve sparing (modified template alone: early 75–87%, refined 97%).
- **Also know:** ileus (commonest); chylous ascites; bleomycin lung toxicity (low FiO_2); nephrectomy needed at PC-RPLND (left-sided mass >10 cm); GETUG T93BP showed incomplete RPLND predicts relapse.

Recent advances

- **Robotic RPLND** for low-volume disease (same template, same rules).
- RPLND for low-volume stage II seminoma (**SEMS, PRIMETEST**); serum **miR-371a-3p** for selection.

Diagram to draw: great vessels with the three nodal regions, right and left landing zones, and the right-to-left crossover arrow; outline the two modified templates.

Key points to remember:

- Right testis to interaortocaval; left to left para-aortic; crossover only right to left.
- Limits: renal vessels, ureters, ureteric crossing of iliacs; omit suprahilar and interiliac.
- Nerve-sparing of L1–L4 fibres gives antegrade ejaculation >90%.
- PC-RPLND: full bilateral template unless mass ≤5 cm in landing zone.

Q3a. Papillary necrosis.

Type: SAQ (10 marks)

Definition

- **RPN (renal papillary necrosis):** ischaemic necrosis of the renal papillae (inner medulla), which may slough into the collecting system.

Pathophysiology

- The papilla is vulnerable: it depends on the **vasa recta**, has **low oxygen tension** and a **hypertonic** environment. Almost every cause reduces medullary blood flow or oxygenation.
- **Sickle cell:** hypoxia and hypertonicity make cells sickle and block the vasa recta, causing microinfarcts.
- **Analgesics:** NSAIDs (non-steroidal anti-inflammatory drugs) block vasodilator prostaglandins; metabolites concentrate in the medulla.
- **Diabetes:** microangiopathy plus infection.

Causes ("POSTCARDS")

Letter	Cause
P	Pyelonephritis
O	Obstruction
S	Sickle cell disease or trait
T	Tuberculosis
C	Cirrhosis
A	Analgesic abuse
R	Renal transplant rejection; renal vein thrombosis
D	**Diabetes mellitus**
S	Systemic vasculitis

- **Also know:** infant dehydration and hypoxia; shock; renal candidiasis.

Clinical features

- **Acute:** fever, flank pain, haematuria, **ureteric colic from a sloughed papilla**, sepsis; bilateral cases may cause AKI (acute kidney injury).
- **Chronic or incidental:** polyuria (poor concentration), sterile pyuria, recurrent UTI (urinary tract infection), CKD (chronic kidney disease). Retained papillae may **calcify** and harbour infection, including fungi.

Investigations

- Urine microscopy and culture; **strain urine** and send tissue for histology; creatinine, glucose, haemoglobin electrophoresis.
- **CT urography or IVU (intravenous urography):** early contrast tracking into the papilla ("**ball-on-tee**", "**lobster claw**"); "**ring sign**" (contrast around a detached papilla); clubbed calyces; filling defects; triangular ring-shaped calcification.
- Retrograde pyelography when renal function is poor.
- **Differential diagnosis:** tuberculosis, medullary sponge kidney, calyceal diverticulum, upper tract urothelial carcinoma.

Management

- **Treat the cause:** stop analgesics, control diabetes, hydrate, manage sickle crisis.
- **Culture-guided antibiotics.**
- **Obstruction by a sloughed papilla:** urgent drainage (stent or nephrostomy), then ureteroscopic extraction.
- Remove calcified retained papillae that cause persistent infection.
- **Follow-up:** renal function; analgesic abusers risk urothelial cancer.

Diagram to draw: calyx showing ring sign and lobster-claw appearance.

> **Key points to remember:**
>
> - Vasa recta, low oxygen and hypertonicity make the papilla vulnerable.
> - POSTCARDS; diabetes, analgesics, sickle cell, obstruction with infection.
> - Ring sign; sloughed papilla causes colic and obstruction: drain urgently.

Q3b. MAG 3

Type: SAQ (10 marks)

Assumed meaning: Tc-99m MAG3 (mercaptoacetyltriglycine) renal scan: principle, technique, interpretation and uses.

Principle

- **^{99m}Tc-MAG3:** protein-bound tracer cleared mainly by **proximal tubular secretion**. Extraction is nearly **60%** per pass, versus about 20% for DTPA (diethylenetriaminepentaacetic acid).
- So it gives **good images even with poor GFR (glomerular filtration rate)**, lower background and lower cost. **Agent of choice for split function and obstruction.**

	DTPA	MAG3	DMSA (dimercaptosuccinic acid)
Handling	Glomerular filtration	Tubular secretion	Tubular cortical binding
Study	Dynamic	Dynamic	Static (2 h)
Key use	GFR; renovascular	Split function, drainage	Scars, pyelonephritis

Technique ("well-tempered" diuretic renogram)

- **Hydration**; **bladder catheter** in children and neurogenic bladder.
- Dose 185–370 MBq (5–10 mCi) intravenously in adults.
- **Furosemide:** 0.5 mg/kg (maximum 40 mg) in adults; 1 mg/kg in children, given when the system is full. Timing varies (F+15, F0, F−15), so the report must state it.
- Upright or post-void images test gravity drainage.

Interpretation

- **Perfusion phase** (first minute): renal curve follows aortic upstroke.
- **Uptake phase: DRF (differential renal function)** from area under curve at 1–3 minutes; **normal 45–55%**. Normal peak at 3–5 minutes, bladder by 10–15 minutes.
- **Drainage phase:** post-furosemide T½ (half-clearance time):

T½	Meaning
<10 min	Not obstructed
10–20 min	Indeterminate
>20 min	Obstructed

- **Curve types (O'Reilly):** I normal; II rising (obstructed); IIIa dilated, washes out with diuretic; IIIb equivocal.

Uses

- PUJ (pelviureteric junction) obstruction and megaureter: **surgery if DRF <40% with poor drainage or falling DRF**; follow-up after pyeloplasty.
- Transplant: acute rejection (perfusion falls) versus delayed graft function (perfusion kept); urine leak.
- **Also know:** indirect cystography for VUR (vesicoureteric reflux); split function before nephrectomy.

Pitfalls

- Dehydration, full bladder, wrong ROI (region of interest), mistimed diuretic, massive dilatation and neonatal immaturity mimic obstruction. **Trend on serial scans** matters.

Diagram to draw: renogram curves types I, II, IIIa and IIIb.

Key points to remember:

- MAG3: tubular secretion, about 60% extraction; best for split function and drainage.
- DRF normal 45–55%; T½ >20 min obstructed, <10 min not.
- Hydration, catheter, correct diuretic timing: the well-tempered renogram.

Q3c. Hari Asopa and Charles Huggins

Type: SAQ (10 marks)

Assumed meaning: contributions of each to urology.

Hari S. Asopa (Indian urologist)

1. Asopa dorsal inlay urethroplasty (2001)

- Single-stage graft urethroplasty for anterior urethral stricture by a **ventral sagittal urethrotomy** approach.
- **Steps:** expose the urethra ventrally without circumferential mobilisation; open the urethra ventrally through the stricture; incise the **dorsal urethral plate** in the midline down to the corpora; **inlay** a free graft (buccal mucosa or penile/preputial skin) and quilt it to the tunica albuginea; close the urethra over a catheter.
- **Rationale:** the graft lies on the well-vascularised corporal bed; no mobilisation preserves spongiosal blood supply; suits bulbar and penile strictures.
- Ranks with **Barbagli dorsal onlay** and **Kulkarni one-sided** repair among dorsal graft techniques.

2. Asopa hypospadias repair (1970s)

- One-stage repair using a **vascularised inner preputial island flap** still attached to the outer preputial skin, transferred ventrally to form the neourethra; a forerunner of the Duckett transverse preputial island flap.

Charles B. Huggins (1901–1997)

- Canadian-born American urologist, University of Chicago.
- **1941, with Clarence Hodges:** castration or oestrogen caused **regression of metastatic prostate cancer**, with falling serum acid phosphatase, pain relief and weight gain; injected androgens made it worse. This proved that **prostate cancer is hormone (androgen) dependent**.
- Used **serum acid phosphatase** as the first tumour marker to monitor response.
- **Nobel Prize in Physiology or Medicine, 1966** (shared with Peyton Rous) for hormonal treatment of prostate cancer.
- **1945:** bilateral adrenalectomy for relapse after castration, suspecting adrenal androgens; responses brief, morbidity high.
- **Also know:** canine prostate physiology; adrenalectomy for breast cancer.

Legacy

- Basis of all ADT (androgen deprivation therapy): orchidectomy, oestrogens, LHRH (luteinising hormone-releasing hormone) agonists and antagonists, and later abiraterone (blocking adrenal androgens).

Diagram to draw: dorsal inlay: urethra opened ventrally, dorsal plate incised, graft on corpora (cross-section).

> **Key points to remember:**
>
> - Asopa: ventral sagittal urethrotomy, dorsal plate incised, free graft inlaid (2001).
> - Huggins and Hodges 1941: castration regresses prostate cancer; Nobel 1966.
> - Huggins 1945: adrenalectomy for castration-resistant disease.

Q3d. Genetic counselling

Type: SAQ (10 marks)

Assumed meaning: genetic counselling in urological practice.

Definition

- A communication process that helps patients and families understand the **medical, psychological and family implications** of a genetic condition, so they can make informed choices. It is **non-directive, voluntary and confidential**.
- **Germline testing** (blood) finds inherited variants with family implications; **somatic testing** (tumour) guides treatment.

Steps

1. **Three-generation pedigree**.
2. **Risk assessment:** inheritance and recurrence risk (autosomal dominant 50%; autosomal recessive 25%).
3. **Pre-test counselling:** which test; possible results (pathogenic, **VUS [variant of uncertain significance]**, negative); insurance and psychological harms; consent.
4. **Post-test counselling:** interpret (do not act on a VUS); **cascade testing of relatives**; surveillance; targeted therapy.
5. **Reproductive options:** prenatal diagnosis, PGT (preimplantation genetic testing), donor gametes; psychosocial support.

Urological applications

Condition	Genes	Key point
Prostate cancer	BRCA2 (> BRCA1), ATM, CHEK2, PALB2, mismatch repair (Lynch), HOXB13	NCCN (National Comprehensive Cancer Network): test metastatic, high-risk, strong family history; BRCA2: aggressive disease, PARP (poly-ADP-ribose polymerase) inhibitors, earlier screening
Hereditary RCC (renal cell carcinoma)	VHL, MET, FH, FLCN, TSC1/2, SDH, BAP1	Counsel RCC at ≤46 years, bilateral or multifocal, family history
UTUC (upper tract urothelial carcinoma)	Lynch (MSH2 highest)	Screen if <60 years or Lynch-spectrum cancer history
ADPKD (autosomal dominant polycystic kidney disease)	PKD1 (chromosome 16), PKD2 (chromosome 4)	Testing selects living related donors
Cystinuria	SLC3A1, SLC7A9 (recessive)	Screen siblings
Male infertility	Karyotype, AZF (azoospermia factor) deletion, CFTR (cystic fibrosis transmembrane conductance regulator)	AZF deletions pass to sons by ICSI (intracytoplasmic sperm injection); absent vasa: test partner for CFTR
DSD (differences of sex development)	e.g. CAH (congenital adrenal hyperplasia, recessive), AIS (androgen insensitivity, X-linked)	Recurrence and gonadal tumour risk

Ethics and law

- Confidentiality; right not to know.
- **India:** PCPNDT Act (Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994) bans disclosure of fetal sex.

Diagram to draw: three-generation pedigree with standard symbols.

> **Key points to remember:**
>
> - Non-directive; pre-test and post-test counselling; cascade testing.
> - Prostate: BRCA2 most important; test all metastatic men.
> - RCC ≤46 years, ADPKD, cystinuria, AZF/CFTR, DSD are classic urological indications.

Q3e. Therapeutic angioembolization.

Type: SAQ (10 marks)

Definition

- Deliberate **endovascular occlusion** of a target vessel, usually by transfemoral selective catheterisation under fluoroscopy, to stop bleeding, devascularise a tumour or close an abnormal vessel.
- **Superselective** embolisation of a segmental branch preserves most parenchyma.

Embolic agents

Type	Examples	Use
Temporary	Autologous clot; gelatin sponge (2–6 weeks)	Priapism, prostate, bladder bleeding
Permanent particles	PVA (polyvinyl alcohol), microspheres	Tumours, PAE (prostate artery embolisation)
Mechanical	Microcoils, vascular plugs	Pseudoaneurysm, AVF (arteriovenous fistula), varicocele
Liquid	NBCA (N-butyl cyanoacrylate) glue, absolute ethanol	Tumour ablation, recurrent priapism

Urological indications

- **Renal trauma:** active contrast extravasation, pseudoaneurysm, AVF; success high in grade IV, lower in grade V. Failure predictors: instability, grade V, urinary extravasation. Repeat embolisation often succeeds.
- **Iatrogenic bleeding** after PCNL (percutaneous nephrolithotomy) or partial nephrectomy: **treatment of choice**; superselective coils nearly 100% technically successful.
- **AML (angiomyolipoma): first line for acute haemorrhage**; prophylactic for >6 cm, aneurysm >5 mm, growth or pregnancy plans. In TSC (tuberous sclerosis complex), embolise then give steroids.
- **RCC:** palliation of haematuria or pain in unfit patients; preoperative use no longer routine.
- **Intractable bladder haemorrhage** (radiation cystitis, cancer): bilateral **anterior division of internal iliac**; avoid the posterior division (gluteal pain).
- **Prostate:** PAE for BPH (benign prostatic hyperplasia) and refractory prostatic haematuria.
- **High-flow priapism:** superselective cavernosal artery embolisation; resolution ≥75%; recurrence → repeat with a non-resorbable agent.
- **Also know:** varicocele (venous coils); lymphatic embolisation for chylous ascites; ADPKD (polycystic kidney) pain.

Complications

- **Post-embolisation syndrome:** fever, flank pain, nausea, leucocytosis; up to 72 hours; self-limiting.
- **Non-target embolisation:** bowel, bladder, rectum, gluteal muscle, penile skin.
- Renal infarction or abscess; contrast nephropathy; access-site haematoma; coil migration; ED (erectile dysfunction) after priapism embolisation.

Requirements

- Experienced team, resuscitation in the suite; unstable non-responders need surgery.

Diagram to draw: renal angiogram showing segmental pseudoaneurysm before and after coil occlusion.

Key points to remember:

- Superselective coils: treatment of choice for pseudoaneurysm and AVF.
- First line for bleeding AML; prophylactic >6 cm or aneurysm >5 mm.
- Bladder bleeding: anterior division of internal iliac.
- Post-embolisation syndrome is common and self-limiting.

Q3f. Endorectal coil MRI

Type: SAQ (10 marks)

Principle

- **ERC (endorectal coil):** a receive-only surface coil inside an inflatable balloon placed in the rectum against the prostate, used with an external pelvic phased-array coil.
- **"The currency of MRI is signal."** A coil close to the prostate gives higher **SNR (signal-to-noise ratio)**, allowing **higher spatial resolution**.

Technique

- Antispasmodic to reduce rectal motion artefact; balloon filled with air or perfluorocarbon (fluid reduces susceptibility artefact).
- Sequences as for mpMRI (multiparametric MRI): **T2W (T2-weighted)** triplanar for anatomy, **DWI (diffusion-weighted imaging) with ADC (apparent diffusion coefficient)** map, **DCE (dynamic contrast-enhanced)**; ERC was traditionally needed for **MRSI (MR spectroscopic imaging)**.

Uses

- **Local staging:** extraprostatic extension and seminal vesicle invasion (both low T2 signal) before radical prostatectomy.
- **Local recurrence** after prostatectomy: recurrence isointense on T1, slightly hyperintense on T2; DCE raises accuracy at low PSA (prostate-specific antigen).
- **Also know:** focal therapy planning (ERC recommended at 1.5 T); seminal vesicles; female urethral diverticulum.

ERC versus no ERC

	With ERC	Without ERC (3 T, pelvic coil)
Signal and resolution	Higher	Adequate at 3 T
csPCa (clinically significant cancer) detection	GG (Grade Group) ≥3 equal; GG2 slightly better	GG ≥3 equal
Comfort, cost, time	Worse	Better
Gland	Deformed (fusion and radiotherapy planning harder)	Natural shape

Disadvantages

- Artefacts: **near-field signal flare** next to the coil (needs intensity correction), air susceptibility at the rectal wall, malposition.
- **Contraindications:** absent rectum (abdominoperineal resection), recent rectal surgery, anorectal fistula or inflammatory bowel disease, severe haemorrhoids or stricture, latex allergy.

Current position

- At **3 T the ERC is largely historical**; PI-RADS (Prostate Imaging Reporting and Data System) v2.1 does not require it. **Consider it at 1.5 T** or with older scanners, or when body-coil quality is poor.

Diagram to draw: sagittal pelvis with inflated endorectal coil against the posterior prostate.

Key points to remember:

- Coil close to prostate: higher SNR and resolution.
- Useful at 1.5 T; largely unnecessary at 3 T (GG ≥3 detection equal).
- Drawbacks: discomfort, gland deformation, near-field flare.

Rapid Revision Sheet – Practice Set 7 Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Prune Belly Syndrome. Antenatal diagnosis, pathophysiology and postnatal management. (LAQ)

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- Right testis to interaortocaval; left to left para-aortic; crossover only right to left.
- Limits: renal vessels, ureters, ureteric crossing of iliacs; omit suprahilar and interiliac.
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- Vasa recta, low oxygen and hypertonicity make the papilla vulnerable.
- POSTCARDS; diabetes, analgesics, sickle cell, obstruction with infection.
- Ring sign; sloughed papilla causes colic and obstruction: drain urgently.

Q3b. MAG 3 (SAQ)

- MAG3: tubular secretion, about 60% extraction; best for split function and drainage.
- DRF normal 45–55%; T½ >20 min obstructed, <10 min not.
- Hydration, catheter, correct diuretic timing: the well-tempered renogram.

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- Non-directive; pre-test and post-test counselling; cascade testing.
- Prostate: BRCA2 most important; test all metastatic men.
- RCC ≤46 years, ADPKD, cystinuria, AZF/CFTR, DSD are classic urological indications.

Q3e. Therapeutic angioembolization. (SAQ)

- Superselective coils: treatment of choice for pseudoaneurysm and AVF.
- First line for bleeding AML; prophylactic >6 cm or aneurysm >5 mm.
- Bladder bleeding: anterior division of internal iliac.
- Post-embolisation syndrome is common and self-limiting.

Q3f. Endorectal coil MRI (SAQ)

- Coil close to prostate: higher SNR and resolution.
- Useful at 1.5 T; largely unnecessary at 3 T (GG ≥3 detection equal).
- Drawbacks: discomfort, gland deformation, near-field flare.